

A Low-Cost First Test of a Brain-Boosting Idea in Rats

Introduction in simple words · Researcher: Solano Moreira Furlan

What is the idea?

Scientists found that a natural substance in the fluid around young brains, called FGF17, seems to help old mice remember things better. Another substance, called 7,8-DHF, is thought to help brain cells make stronger connections with each other. We want to know: if we give both, and also give the animals a rich, stimulating place to learn, do they become better at solving different kinds of problems than normal rats of the same age?

What will we do?

We will use a small number of healthy young rats (64 in the main test, 136 counting helper animals) in four groups: normal rats; rats living in the learning environment; rats given the two substances; and rats given the two substances and the learning environment. Later we test all of them on three different tasks — a puzzle where the rules suddenly change, a maze with new problems, and a swimming-pool memory test. Before the main test, a small first step checks whether FGF17 switches on its expected signal in the rat brain (we look at 7,8-DHF too, but only descriptively; this does not prove either substance reaches every brain cell). If that first signal is missing, we stop early and save money. Some scientists also doubt that 7,8-DHF works the way people first thought, so we treat that as an open question. The rats get a small brain operation, daily injections, limited food for several weeks and swimming tests, and are humanely put to sleep at the end; an animal-ethics committee must approve everything first.

Who does the work?

A good laboratory in India will run the whole experiment, using this document together with detailed lab procedures that the lab staff write and the researcher approves. The researcher guides it from far away and keeps the secret group codes, so the staff giving the medicines do not know which rat got what (they can see which rats live in the enriched space), and the researcher checks the results independently.

How much will it cost?

The first step and the main experiment together are planned to cost about USD 10,200 for materials only (supplies, animals, and chemicals). It could be as low as 8,900 or as high as 12,200; the researcher has approved a ceiling of USD 12,000 for materials. Staff pay and the lab's overhead charges are NOT included, and together they could be a large part of the real bill; the Indian lab must quote them. A second part (looking at the brain tissue under the microscope) costs about USD 1,700 more and fits inside the ceiling at the expected prices but not if prices run high, so it is done last and dropped first.

What will this test tell us — and not tell us?

This is only a first, small test. It can tell us whether the experiment is practical, whether FGF17 switches on its expected signal in the brain, and whether the results lean in a promising direction (with so few animals, a real but modest effect would often be missed). It cannot prove that the idea works, and it cannot say anything about people.

What if it goes well?

Then we plan a bigger, stronger experiment. Its first part would separate what each ingredient does. Its second part would use optogenetics — a new technique that switches chosen brain cells on and off with light — to see how brain activity interacts with these substances.

The long-term dream

In the far future we hope this kind of approach might help people who have learning difficulties or problems with how brain cells connect, and older people with Alzheimer's disease, by helping the brain rebuild connections. That is a hope, not a result. Many steps, long safety testing, and strict ethics rules come first, and there are real obstacles (for example, how to get these substances into a human brain safely). The rest of this document gives the technical details.

Technical protocol (India; core consumables ≈ USD 10.2k, range 8.9–12.2k)

Does a candidate plasticity-promoting treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive tasks in rats? Staged feasibility study run by an Indian host laboratory under remote oversight.

Researcher / sponsor	Solano Moreira Furlan (remote; the host laboratory runs the experiment)
Document	India initial-test protocol, version 10 (final; rounds 1–8 applied, then a full read-through audit)
Date	9 October 2026
Status	Not approved by an ethics committee. Every reviewer verdict so far: “approve with changes”. Prices are estimates, not quotes. “crf” in the request was read as CSF (assumption to confirm).

Executive summary (technical)

Question. Does a candidate plasticity-promoting treatment — FGF17 (identified in young cerebrospinal fluid; Iram 2022 [1]) plus 7,8-DHF (a putative TrkB/BDNF-pathway agonist whose mechanism is disputed [2–4]) — combined with a structured enriched learning environment (ELE) raise the performance of healthy young rats across three tasks, one of which (set-shifting) uses stimulus dimensions not practised in the ELE, compared with untreated peers?

Design. 64 rats in the main 2×2 (≈10–11 completers per arm); 136 rats in total including 48 engagement satellites. Staged: Phase 0a (engagement and reagents) must pass before Phase 0b (behaviour); Phase 0c (EdU/Olig2 histology) is a deferred add-on on banked tissue. It informs, and does not confirm, whether a larger study is justified.

Budget (honest). Consumables-only core (0a + 0b, tissue banked) ≈ USD 10.2k base; ≈ 8.9k optimistic; ≈ 12.2k pessimistic. Phase 0c adds ≈ 1.7k. The researcher has approved a materials ceiling of USD 12k: the core fits in the base and optimistic cases and slightly exceeds it in the pessimistic case; core plus 0c fits at base prices (11.9k) but not in the pessimistic case (14.1k), so 0c is run last and stops if prices run high. Personnel, service fees and institutional overhead are not included and must be quoted by the host; the all-in cost is unknown until then.

1. Aim, hypotheses and what the data can claim

- **Primary aim (feasibility):** estimate completion rate, variance and direction of performance in a 2x2 of drug package × environment; establish reagent activity and pathway engagement.
- **Exploratory hypothesis:** D+E (drug package + enrichment) performs better than control peers (C) and than either component alone.
- **Transfer:** set-shifting is the primary transfer readout (odour and digging-medium dimensions are excluded from the ELE). Hebb-Williams and the water maze are spatial; the ELE has tunnels, climbing and wheels, so they are flagged as spatial, not transfer.
- **Allowed claims:** feasibility, attrition, variance, direction with 95% interval of D+E – C (overall index; set-shifting reported separately), pathway engagement (ERK), descriptive structure if Phase 0c is run — for the drug package as a whole.
- **Not allowed:** “smarter”, “general intelligence” or a g factor [8,9]; neuroplasticity; single-agent claims; TrkB-specific claims (pERK shows pathway engagement only); human or disease claims; replication of Iram 2022 (aged-mouse rescue; this asks about enhancement in healthy young adults, which no study found has tested).

2. Long-term goal, next experiment (including optogenetics) and translational roadmap

Recommended wording (translational reviewer, adopted):

“Our long-term aim is to test whether combining a putatively plasticity-promoting biological treatment (an FGF17-type factor plus a validated TrkB/BDNF-pathway agent) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer’s disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Prior neurotrophin programmes in Alzheimer’s have not shown benefit (including a direct-delivery AAV-NGF trial), brain delivery of protein factors is a major obstacle, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates.”

Stage	Content	Gate to proceed
0 – this study	0a engagement/reagents → 0b behaviour in healthy young rats → 0c histology (deferred add-on).	Section 8.
1a – next experiment: powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment (and exercise); sustained infusion by pump; dose-response; durable vs on-drug effects; TrkB and FGFR necessity tests; synaptic readouts (PSD-95/synaptophysin on banked tissue, LTP).	Replicated effect on a pre-registered primary; structural effect.
1b – next experiment: optogenetics (only if 1a is positive; mechanistic, not translational)	Question: how does controlled neuronal activity interact with FGF17 and with 7,8-DHF-driven plasticity? Candidate designs (to be designed and reviewed in their own round): (a) activity × plasticity-opening — optogenetic stimulation of a defined circuit (e.g. CaMKIIa-driven channelrhodopsin in prefrontal or hippocampal pyramidal neurons) during or after learning, with and without FGF17/7,8-DHF, asking whether activity amplifies oligodendrocyte-precursor proliferation, myelination and synaptic plasticity (mouse precedent: premotor-cortex stimulation, Gibson 2014 [10]; extension to prefrontal/hippocampal circuits is an extrapolation); (b) necessity — inhibitory opsins during learning or consolidation; (c) cell-type targeting of oligodendrocyte-lineage cells needs Cre lines that are mostly mouse (species switch or an AAV-based rat strategy). Requirements: AAV vectors with institutional biosafety (IBSC) approval (referral to RCGM only if the risk category requires), fibre implants, calibrated light sources, opsin-negative/fluorophore-only and light-only controls (heating, visual artefacts), histological verification of opsin and fibre placement, and monitoring for stimulation-induced seizures/kindling. Beyond this budget; own funding, equipment, expertise and ethics approval.	1a positive; opsin tools validated in the host lab.
2 – disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer’s models (5xFAD, APP/PS1, TgF344-AD) — amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models at appropriate ages: Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded) as separate tracks.	Benefit with engagement and structural evidence.

Stage	Content	Gate to proceed
3 – drug development	Clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — unproven); a validated TrkB-pathway drug with human-relevant PK; two-species toxicology; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (CDSCO NDCT Rules 2019; ICMR ethics; US/EU IND).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, cognitive tests); structured learning input as digital training; Phase 2 in MCI/early AD and, separately, in neurodevelopmental populations (DCGI scrutiny; assent plus proxy consent; independent data-safety monitoring).	Safety and biomarker signal.

Hard barriers that must be stated: no validated way exists to deliver FGF17 to the human brain; the mitogenic and angiogenic risk of an FGF-class CNS agent may never clear a paediatric risk-benefit assessment for a non-fatal indication such as a learning disability, and enhancement in people without disease carries a higher ethical bar than treatment of disease; 7,8-DHF is reported not to activate TrkB reliably in cell assays [2–4]; no human trials of FGF17 or 7,8-DHF were found (verify); healthy-animal enhancement may not transfer to disease or humans. The FGF23/phosphate effect is not the relevant class risk (a different, endocrine FGF).

What this study must deliver to earn Stage 1a: reagent activity and FGF17 pathway engagement (0a); a positive direction on the overall index (0b), with set-shifting reported separately; and, if Phase 0c is run, an EdU+/Olig2+ direction. Covariation of structure with behaviour cannot be judged at this n.

3. Design and animals

Young adult male Wistar rats (about 8–10 weeks old, ≈ 250 – 300 g at surgery) from a CPCSEA-registered source (the regulator is now also called CCSEA; ICAR-IVRI Izatnagar INR 350; NISER INR 300–500 plus transport; NIN's animal unit is being reorganised — confirm supply directly). Males only. Housed on a reversed 12:12 light cycle so that dark-phase work falls in working hours. The sponsor randomises by baseline weight and source litter and holds the allocation and drug-code key (section 10); **each wave contains 8 rats per arm**; coded syringes/vials; surgeon, injector and behaviour operators blind to drug. The environment arm cannot be fully blinded, so ELE handlers and test-day scorers are different people, all behaviour is video-recorded under coded IDs and scored blind, 20% are re-scored for inter-rater agreement (ICC), and apparatus is cleaned between rats. Slots are balanced across arms (one rat per arm per slot), so arm is not confounded with time of day.

Group	Content	Enrolled
Main cohort	4 arms $\times$ 16: C (vehicle + standard), E (vehicle + ELE), D (FGF17 + 7,8-DHF + standard), D+E; 2 waves $\times$ 8 per arm	64
Engagement satellite – FGF17	Time-matched vehicle (4 each) and FGF17 (6 each) at 15 / 30 / 60 min after an ICV bolus ($0.5 \mu\text{g}$ in $3 \mu\text{l}$, as in the main cohort), restraint duration logged; rapid dissection and snap-freezing: a ≈ 1 -mm micropunch of the periventricular wall, which necessarily includes a thin rim of adjacent tissue (the claim to exclude striatum and septum is dropped); μg of protein per punch is measured by BCA in the practice group, which fixes the punch size (1 – 1.5 mm), the lane and replicate budget, and whether a band is quantifiable; and hippocampus; pERK/total ERK	30
Engagement satellite – DHF	Vehicle / 45 min / 120 min after the 7th daily dose (6 each); pERK/total ERK (descriptive)	18
Partner rats	12 unoperated partners (6 for ELE arenas, 6 for control arenas; never swapped; aggression pre-screened; ≈ 5 rats per partner per condition; ≈ 2.7 pair-sessions per partner per day, logged)	12
Practice/pilot	6 cannula/dye placement and EdU labelling checks (incl. storage-time control); 6 vehicle-tolerability checks (3 low-DMSO solution vs 3 CMC-Na suspension, 28-day interim gate then continued to day 53, with weekly weight and injection-site scoring, necropsy scoring and a pre-specified switch rule: $>10\%$ weight loss or adhesion score $\geq 2 \rightarrow$ switch vehicle)	12
Total	Cannulated rats: $64 + 30 + 6 = 100$	136

Enrolment is 16 per arm because ICV-cannula attrition in rats is commonly 30–40%; the plan assumes 62–69% completion (10–11 per arm). Cannula loss is logged by arm. Because climbing, wheels and partners exist only in the enrichment arms, cap loss is expected to be higher there: protected recessed caps with a smooth collar, climbing structures ≥ 15 cm below the lid and solid-surface (rung-free) wheels are used, an arm-stratified tipping-point analysis is pre-specified, the delivered FGF17 dose is reported per D+E completer, and a completion difference between arms of >20 percentage points is flagged and >25 triggers a review (and possible stop). **Welfare:** the combined burden (cannula for ≈ 10 weeks, 53 daily injections, food restriction, swimming) is classified as moderate severity, with a scoresheet and humane endpoints ($>15\%$ weight loss, weight below 85% of the growth-adjusted target, injection-site lesions, cannula infection, hypothermia rule); the IAEC will be asked for a pre-submission review of severity. Pre-specified exclusions: dye/histology-confirmed ventricular placement, humane endpoints, swim speed $<50\%$ of cohort median. Housing: pair-housed with a perforated divider if the IAEC permits, otherwise single-housed, identically in all arms.

Power (honest). With $N=40$ – 44 completers an interaction contrast reaches 80% power only for $f \approx 0.43$ – 0.46 , and a pairwise D+E vs C comparison only for $d \approx 1.3$ (unadjusted); power at $f=0.25$ is 0.34 – 0.37 . A cross-species enrichment meta-analysis reports $g \approx 0.44$ in rodents (an unpublished dissertation [15]), and healthy young rats are near ceiling on simple tasks, so a null result is likely even if a modest true effect exists. Pilot effect sizes must not be used to power the next experiment from the point estimate [12,13].

4. Interventions

Component	Specification
FGF17 (days 0–6)	Recombinant mouse FGF17 (mouse–rat sequence alignment documented before ordering). Order one 50 µg pack first, bioassay it (≈10 µg), then order the second 50 µg pack (second import, 6–12 weeks, is on the critical path). Chronic guide cannula to the right lateral ventricle (scaled coordinates). Four boluses on days 0, 2, 4, 6: 3 µl of ≈167 µg/ml (0.5 µg each; 2.0 µg per rat) in aCSF + 0.1% BSA (standard anti-adsorption practice, not a vendor specification) + heparin 10 µg/ml, awake, 0.5–1 µl/min with the injector left 2–3 min; aliquots thawed fresh each session; the 167 µg/ml working stock lies within vendor reconstitution ranges (typically ≥100 µg/ml, up to 1 mg/ml; reconstituted FGF17 keeps 2–7 days at 4 °C, aliquots about 3 months at –20 °C), so recovery after reconstitution and after freeze-thaw is still checked (activity or ELISA), low-bind tubes are used, the solution is inspected for precipitate, and if unstable the concentration is lowered and the volume raised (up to 5 µl; if 100 µg/ml also precipitates the delivered mass falls below 0.5 µg and the shortfall is declared); injector and line dead volume are measured and primed so that 3 µl is delivered. Supply: ≈73 µg of doses (32 × 2.0 + 18 × 0.5) plus ≈25 µg for bioassay (≈10 µg), dead volume and repeats (≈15 µg) = ≈100 µg (a budget option of 0.3 µg boluses, 1.2 µg total, saves ≈0.3k but gives only ≈0.6× the infused mass). Re-test animals (12 rats, 6 per group, contingency only) are registered as a maximum. Vehicle boluses in all other rats. <u>Delivery gap and dose anchor</u> : Iram 2022 infused continuously for about a week with an Alzet 1007D (≈84 µl delivered); one reviewer retrieved the Methods as recombinant mouse FGF17 at 25 µg/ml (≈2.1 µg continuous) plus a separate acute 0.5 µg bolus in 2 µl — another reviewer could not retrieve it, so the host must confirm it in the Methods (PMC9377328 or the UCL copy) before ordering. Four 0.5 µg boluses (2.0 µg total, 12 µl) match the infused mass but are pulsatile and ≈7× smaller in volume; a null is therefore still confounded with delivery and FGF17 stays exploratory.
7,8-DHF	5 mg/kg IP daily, days 0–52 (53 injections; rotated sites). Daily order: testing (if scheduled) → 7,8-DHF injection → ELE/control session 30 min later, so the agent is on board during enrichment (mouse half-life ≈2 h) and the next day's testing falls at trough (≈18–24 h after dosing in mice; one rat report gives a terminal half-life of ≈23 h, route unclear, so the trough assumption is uncertain in rats). Each rat has a fixed daily time slot (test → injection → ELE/control 30 min later) so its dosing-to-test interval stays constant. There are 8 slots per day; each slot holds 4 rats, one from each arm (C and D in the two control arenas, E and D+E in the two ELE arenas), so arm is balanced against time of day without rotation; slot assignment is random within arm. Rat precedent for 5 mg/kg IP daily includes 5 weeks in a rotenone model in 9-month-old rats [17] and a 1%-DMSO-in-saline regimen in an optic-nerve model (outcomes at 7–21 days) [18]; mouse work exists [5]. The 53-day duration in healthy young rats therefore goes beyond direct precedent (the longest identified is 5 weeks). Vehicle (decided by the tolerability check, 6 rats: 3 low-DMSO solution, 3 CMC-Na suspension; weekly weights, injection-site scoring and necropsy; the 28-day interim reading fixes the vehicle before wave 1 is ordered, and dosing continues to day 53 with the same observations; a problem found after day 28 pauses wave 1 before its day 0): default is a low-DMSO solution (5% DMSO / 40% PEG300 / 5% Tween-80 / 50% water per Selleck, or 10% / 40% / 5% / 45% saline per MedChemExpress/InvivoChem, ≈2.5 mg/ml; prepared fresh); fallback is an aqueous CMC-Na suspension (no peer-reviewed intraperitoneal PK found; an IP suspension of an insoluble compound may give erratic, depot-like absorption, so a null could reflect exposure and a plasma/brain bridging check is then required). Oil vehicles are avoided (repeated intraperitoneal oil, especially peanut or mineral, can cause granulomatous inflammation; mild with corn or olive oil in a 5-day mouse study). Necropsy scoring of peritoneal adhesions/granulomas in all main-cohort rats; identical vehicle in all arms.
Enriched learning environment (ELE)	Days 0–52, one 60-min pair session/day (test rat + partner) 30 min after the 7,8-DHF injection, in the dark phase under red light [6,7]. This is a weak enrichment dose (declared). Minimum specification the host must meet regardless of local habit: arena ≥1 m², ≥3 novel objects per session, ≥2 running wheels, ≥3 puzzle devices on the six-level ladder, the same partner for the whole study, session minutes and ladder level logged per rat. SOP with photos to be completed by the host before IAEC submission (arena size, object list ≥3 new objects/day, latch/string-pull/lever/chew puzzle devices on a six-level ladder, running wheels, tunnels/climbing, small fixed-calorie rewards deducted from the daily ration; engagement minutes and ladder progression logged). <u>Excluded from the ELE: odour discrimination and digging puzzles</u> . Control arena: same partner, handling and duration, empty.
EdU (banked label)	Start at 50 mg/kg IP on days 2, 4, 6 (labelling and a stored-tissue control tested in the practice group). Staining is Phase 0c.

5. Timeline and workload (day 0 = first bolus; waves run sequentially)

Days	Activity
–21 to –15	Arrival and 7-day acclimatisation/quarantine; reversed light cycle.
–14 to –11	Cannula surgery with analgesia (≤ 8 rats/day); ≥ 10 days recovery before day 0.
–10 to –1	Recovery, handling, arena habituation, weighing.
0–6	FGF17/vehicle boluses (0, 2, 4, 6); EdU (2, 4, 6).
0–52	ELE/control sessions and 7,8-DHF/vehicle every day (53 doses), weekends included.
12–17	Gradual food restriction from day 12 to 90–95% of the growth-adjusted target by day 17 (before set-shift habituation starts on day 18); held through day 45; humane limit $< 85\%$; daily ration given 2 h after the end of each rat's own ELE/control session (its own slot; on test days the order is test $\rightarrow$ DHF $\rightarrow$ ELE/control $\rightarrow$ ration), with hours since feeding recorded as a covariate.
18–36	Task 1: set-shifting (parameters in Appendix A). Habituation days 18–20, training to criterion days 21–27 (cap 7 days, every rat in its own slot), tests days 28–35: 4 rats (one per arm, from one slot) start per day and are tested on two consecutive days, each rat by its own operator inside its own slot — day 1: SD, CD, ID1; day 2: ID2, ED — with a 3-h cap per rat per day (a stage unfinished at the cap resumes next day; trials are capped at 60 per stage); the overnight gap between day 1 and day 2 is a declared deviation from single-session protocols; the last starters finish on day 36. On most test days 8 rats are in testing (4 on day 1, 4 on day 2), i.e. up to 8 concurrent testers; at least 8 digging boxes are required.
37–45	Task 2: Hebb-Williams — adaptation days 37–39, then 6 chosen problems, one per day (days 40–45), fixed order, 6 trials per problem (3-min cutoff) in two identical mazes run in parallel (arms balanced across mazes); error zones defined in advance; split-half reliability reported. Food restriction ends on the evening of day 45.
46–52	Task 3 (last; non-appetitive): water maze — acquisition days 46–50, probe day 51, visible-platform test day 52 (Appendix A).
53–56	Under anaesthesia an ICV dye injection (3 μ l) confirms patency, then decapitation; a 2-mm coronal slab through the cannula track is cut fresh from the right hemisphere and photographed (dye shows patency; the placement grade is approximate because the cannulated side is not fixed), and the rest is snap-frozen. 8 rats/day balanced by arm and time of day; per-animal handling ≤ 12 min (ice-cold freeze side first). Hemisection: the non-cannulated (left) hemisphere is cut into 1.5–2 mm coronal slabs into 4% PFA for 48 h (counts restricted to cryosections within 1 mm of a cut coronal face, pre-registered), cryoprotected and banked, with a fixation-gradient check on a deep unlabelled section; the remainder of the cannulated (right) hemisphere is snap-frozen and banked at -80°C (dorsal hippocampus and corpus callosum dissected on ice clear of the cannula track, or documented if not; no phospho urgency).

Waves. The two waves are strictly sequential for arena use and surgery: wave 2's day 0 is wave 1's day 71, so wave 2's arrival (its day –21 = wave 1 day 50; acclimatisation only, low load) overlaps wave 1's water maze and euthanasia, and wave 2's surgery (its days –14 to –11 = wave 1 days 57–60) follows wave 1's last euthanasia day. Each wave needs, daily, 32 pair-sessions (16 ELE + 16 control) in 4 arenas = 32 arena-hours (≈ 8 h per arena, ≈ 9 h with changeovers, within the 12-h dark phase), 32 injections and weighing, plus testing, so the working day spans ≈ 14 h (the 12-h dark phase plus set-up and cleaning margins; all testing and sessions fall in the dark phase, and only rations, set-up and cleaning fall in the margins): **ten trained operators in two shifts of five, a surgeon and a weekend rota** (twelve if all testers and arena staff must overlap; if fewer are available, set-shift tests run 2 rats/day over 16 days, which delays Hebb-Williams and the water maze and adds ≈ 8 days to each wave; DHF and ELE then continue daily for every rat until its own last water-maze day, and each day's testers are drawn one per arm so exposure duration stays balanced across arms) (time budget in Appendix B; the cost is not in the budget). The day-1 and day-2 set-shift testers are scheduled in different slots wherever possible so that no more than eight testers overlap. In-life per wave 77 days elapsed (78 inclusive; day –21 to day 56); total ≈ 21 weeks (wave 1 day –21 to wave 2 day 56 = 148 days = 21.1 weeks). Pilots and satellites run before the main cohort in Phase 0a (surgery ≤ 8 rats/day on separate days, 6–8 weeks ahead); Phase 0a, including the gate decision, must be finished and written up before wave 1 rats are ordered (about 6 weeks before day –21), and satellite or pilot surgery never shares a day with main-cohort surgery. The fixed task order (set-shifting $\rightarrow$ Hebb-Williams $\rightarrow$ water maze) is declared; ELE and DHF after testing form a post-training consolidation window, declared.

6. Endpoints and tissue

Primary analysis (per task). The three task scores are standardised with the grand mean and the pooled within-arm SD estimated in-sample (no prior data exist), winsorised on residuals at ± 3 , signs aligned. They are analysed in one linear mixed model (task as a repeated factor; random intercept per animal and slot (8 levels) as a random block; task-specific residual variances; arm $\times$ task terms; wave as a fixed effect; baseline weight as the only covariate), using all animals with ≥ 1 task, with Kenward-Roger intervals; software: R (nlme or glmmTMB) or SAS PROC MIXED; on a singular fit the model falls back to a common residual variance, and convergence/boundary diagnostics are reported. The gate statistic is the **index contrast** (mean of the three z-scores, $D+E - C$) in per-task-SD units; set-shifting alone and the index without the water maze are reported separately. Sensitivity analyses: complete cases; unstructured residual covariance; tipping-point (δ -adjusted) analysis for informative dropout (the mixed model is valid under missing-at-random only). Wheel running is zero in two arms and post-restriction weight is post-randomisation, so neither is a primary covariate. Completion by arm and task is reported with an imbalance flag (>20 percentage points). The inter-task correlation matrix is computed within arm from residuals and is descriptive.

Set-shifting score. Criterion 6 consecutive correct, maximum 60 trials per stage; the primary score is $\log(\text{trials to criterion at the extradimensional shift, capped at 60})$, with rats that do not reach it assigned the cap. If more than 20% of rats hit the cap at ED, the rank/censored (Cox) analysis becomes primary instead; the validity check ($ED > ID2$ cohort-wide) always uses a censored comparison. Conditioning on the post-randomisation intradimensional-shift score is a sensitivity analysis only. **Water maze:** mean proximity to the platform in the 24-h probe [21] (start-position correction stated by the host); quadrant time also reported.

Block	Readouts	n
Behaviour	Three tasks; blinded video scoring.	all completers
Banked tissue (every main-cohort animal)	Left-hemisphere slabs (≤ 2 mm thick, which is the real depth control; counts restricted to cryosections within 1 mm of a cut coronal face, pre-registered) for EdU/Olig2 ($\pm$ later CC1; $\pm$ MBP immunostain or western (cheap, on-hypothesis myelin readout) or a FluoroMyelin/Luxol Fast Blue counterstain; corpus callosum primary, dentate/CA1 secondary; counts valid only within matched depth; immersion fixation gives lower, depth-dependent quality than perfusion — declared); right hemisphere frozen (PSD-95/synaptophysin and other westerns deferred to Stage 1a or a funded add-on). Counts are relative (no stereology) and the pulse-chase measures survival of early-born cells, not proliferation. Co-stored EdU+ and no-EdU control blocks; fixed 48 h, then transferred to sucrose cryoprotectant and stored at -20°C (not left for months in dilute formaldehyde), capped at 12 months; chromogenic EdU if fluorescence background is high.	$\approx 10/\text{arm}$
Engagement (satellites, snap-frozen)	pERK/total ERK in periventricular/subependymal tissue (FGF17 gate anchor) and hippocampus; ice-cold dissection (< 2 min), phosphatase inhibitors (NaF, orthovanadate, β -glycerophosphate), FGF2-stimulated positive-control lysate and pre-specified assay CV on every blot; FGF17 at 15, 30 and 60 min; 7,8-DHF at 45 and 120 min after the 7th daily dose (descriptive); every gel carries a bridging control sample and the FGF2-stimulated lysate so fold-changes compare across gels (the 48 satellite samples cannot share one gel); gels are blocked so each spans groups and time points, lane order is randomised, and pERK and total ERK are probed on the same membrane (strip and re-probe; stripping is validated, otherwise total ERK runs on a parallel gel because stripping can raise CV). With a western CV of ≈ 20 –25% and $n=6$, a 1.5-fold change is near the detection limit for the DHF comparison.	4 vehicle / 6 FGF17 per time point; DHF 6 per group

7. Statistics and pre-registration

- Register on OSF before surgery (sponsor does this); ARRIVE 2.0 [14]; flow diagram of enrolled, excluded and completed animals.
- Estimation only: unadjusted contrasts with Kenward-Roger 95% intervals as primary; an animal-level (cluster) bootstrap that re-runs standardisation, winsorising and the model in every replicate, stratified by arm only, is reported as sensitivity (BCa is unstable at $n \approx 10$); no Holm adjustment because no significance claims are made. Interaction exploratory.
- Partner identity is described only (6 partners per environment cannot support a random effect). Wave (calendar, reagent lots, operator learning) is confounded; arm is balanced across waves so the main contrast is protected, but arm $\times$ wave cannot be assessed with two waves.
- Sizing of the next study: use the upper 80% confidence limit of the pilot's raw-unit residual SD per task [20] and the full factorial/interaction design; the "64 per arm for 0.5 SD" figure applies to a two-group comparison only.

8. Pre-registered decision rules and operating characteristics

Phase	Pass condition			
0a – reagents	FGF17 bioassay active on the first vial; mouse-rat alignment documented; DHF vehicle solution stable (suspension if the fallback is used).			
0a – FGF17 engagement (gate; signalling at the ventricular interface — necessary, not sufficient; the hippocampal punch is the descriptive parenchymal-penetration readout)	periventricular/subependymal pERK/total ERK ≥1.5-fold time-matched vehicle at ≥1 of 15/30/60 min (satellite n = 4 time-matched vehicle and 6 FGF17 per time point; acceptance CV pre-specified; threshold applies to the diluted punch tissue actually assayed, and punches are taken from the right (ipsilateral) hemisphere; positive-control lysate on every blot). Specificity control (added in round 8): a pERK rise over vehicle alone could reflect injection trauma or protein load, so a heat-inactivated FGF17 group (boiled aliquot, equal mass, 4 rats, 30 min, same blots) is registered as 4 additional rats outside the 136 (≈0.1k within the contingency allowance); the gate requires FGF17 ≥1.5-fold over both vehicle and heat-inactivated FGF17. An FGFR-inhibitor arm is deferred to the follow-up experiment. FGF17, an FGF8-subfamily ligand, is expected to signal through c-isoform FGFRs (FGFR3c/FGFR4, per Zhang 2006 [23]; FGF17-specific profile to be verified); oligodendrocyte-lineage cells show ligand-specific FGFR responses, but the cited study tested other FGFs, not FGF17 [16]; periventricular/subependymal receptor expression and the time points are assumptions anchored only by the FGF2 control curve. Simulation at the actual unequal design (4 vehicle / 6 FGF17 per time point, best of three time points; round-8 re-simulation): false-pass ≈6% at assay CV 30% and ≈30% at CV 50%, i.e. about 1.5–2× the earlier balanced 6-vs-6 figures (≈3% and ≈18–19%); true-pass ≈75–93% at a true 1.5-fold change depending on the exact criterion, ≥97% at 2×. Pooling the vehicle animals across time points is allowed only after a pre-test shows no time effect in vehicle pERK. A borderline result (best fold 1.3–1.5) triggers one pre-specified re-test (6 new animals per group at the best time point; 12 contingency rats, registered, outside the 136, plus the 4 heat-inactivated-control rats, i.e. 16 registered rats outside the 136) and the decision then rests on the fresh re-test sample alone. Composite two-stage performance (simulation; lognormal assay noise, effect present at all three time points, re-test = fresh 6 per group at the best time point and pass at ≥1.5-fold): overall pass (balanced 6-vs-6 simulation; the unequal 4-vs-6 design raises false-pass by a similar factor) ≈3% at assay CV 30% and ≈19% at CV 50% if there is no effect; ≈47% and ≈61% at a true 1.3-fold change; ≈85% and ≈82% at 1.5-fold; ≥99% at 2-fold. An independent re-simulation with a looser re-test criterion gave ≈3% and ≈26% under no effect and ≈96–98% at 1.5-fold, so the CV-50% figures depend on that criterion (false-pass about one time in five to one in three); the 82–85% true-pass headline is reproducible only under a stricter, unspecified re-test rule, and the looser-rule figures (≈93%) are the ones that match the stated assumptions. Joint go/no-go percentages in section 8 assume CV 30% and independence between gate and behaviour; at CV 50% the global-null joint rate is higher. A real but modest engagement can be missed. If not met, the main cohort does not start; test a higher dose or continuous infusion in a small follow-up.			
0a – DHF engagement	Reported descriptively, not a gate (n=6, assay CV likely exceeds the effect). If not met, the DHF component is reported as “effect of unestablished mechanism”.			
0b – feasibility	≥50% of the 64 enrolled rats (≥32) complete all three tasks; cannula placement ≥75% correct. (Binomial P(≥32/64): 0.82 at 55%, 0.96 at 60%, 0.995 at 65% true completion; with clustering by wave and surgeon, assuming an intraclass correlation of ≈0.10–0.15, about 0.68, 0.80 and 0.89.)			
0b – direction	Index contrast D+E – C ≥ +0.4 per-task SD, where the SD is the pooled within-arm SD of wave 1 frozen before wave 2 is unblinded (so the gate boundary is fixed in raw task units and the final contrast is re-expressed in those same units; only the coded-group pooled SD is computed between wave 1’s last euthanasia and wave 2’s day 0, treatment identities stay blinded, and the boundary carries ≈±13% uncertainty because the SD rests on ≈32 rats); (illustrative conversion assuming one common inter-task correlation: +0.69 index-SD at r = 0, +0.55 at r = 0.3, +0.47 at r = 0.6) <u>and</u> D+E ≥ both E and D in point estimate. The gate applies to the point estimate; the interval is reported alongside and is not itself a gate.			
Provisional go	Phase 0a passed AND 0b feasibility AND 0b direction (point estimate). Joint probability, illustratively assuming independence, a 0a pass probability of ≈0.9 if FGF17 truly engages and feasibility ≈0.9–0.995: ≈6–7% if FGF17 engages but behaviour is null (≈0.2–2% under a true global null, depending on the gate false-pass rate), ≈50–55% for the target effect. Differential cap loss can select fitter D+E completers and bias the direction upward; the arm-stratified tipping-point analysis addresses this.			
0c (run last; first to drop if prices run high) – confirmed go	EdU+/Olig2+ density in corpus callosum higher in D+E than C (direction). If 0c is not run, the decision stays provisional.			
True situation (8–11 completers per arm; simulated at n=10; index contrast)				
	r=0	r=0.3	r=0.6	r=0.3 + EdU direction*
No effect	5%	8%	10%	4%
Modest (E 0.3, D 0.1, D+E 0.4)	39%	37%	36%	32%
Target (E 0.3, D 0.2, D+E 0.6)	70%	62%	57%	53%
Additive (E 0.3, D 0.3, D+E 0.6)	68%	59%	54%	51%
Large (E 0.5, D 0.3, D+E 1.0)	96%	91%	85%	78%

True situation (8–11 completers per arm; simulated at n=10; index contrast)	r=0	r=0.3	r=0.6	r=0.3 + EdU direction*
Enrichment alone (E 0.5, D+E 0.5)	40%	38%	38%	19%

*EdU direction simulated as independent with a true drug effect of 0.5 SD where a drug effect exists (the “modest” row has a smaller drug effect, so its EdU column is not directly comparable). Other checks: with 8 completers per arm the false-go rate under no effect is ≈10% and the target-scenario go rate ≈59%; the index without the water maze has a ≈9–10% false-go rate. The rule is lenient and **cannot separate synergy from additivity** (62% vs 59%); it only screens for a promising direction. A “go” justifies the larger study only; a “no-go” identifies the failure mode (delivery, engagement, ceiling, attrition) and is not evidence that the approach cannot work.

9. India logistics and budget

Item	What the review found
Ethics	Rats are small animals: IAEC approval is sufficient (no central CPCSEA/CCSEA forwarding). The IAEC must meet at least every 6 months (quarterly for larger programmes); proposals are due ≈30 days ahead; ask for an interim review. The IAEC principal investigator must be host staff.
Imports	FGF17 and antibodies are imported in at least two shipments (vial-first FGF17 order; later orders): 6–12 weeks each, customs-house-agent fees (≈INR 5–15k per consignment), dry-ice courier. Notification 51/96 (as amended in 2000 to include consumables) gave ≈5% duty to DSIR-registered importers; it was consolidated into Notification 45/2025-Customs from 1 Nov 2025, and DSIR certificates may still cite 51/96; whether laboratory reagents count as consumables has been contested before the customs tribunal, so the current position must be confirmed with a customs agent; any GST exemption (2017 notifications 45/2017-CT(R), 47/2017-IGST, against a head-of-institution certificate) is also unconfirmed; the importing institute itself must hold DSIR registration. Without it, landed cost is ≈1.3–1.5× list. Duty/GST rates are reviewers’ knowledge, not verified.
Equipment access (precondition)	The budget assumes written, free access to: stereotaxic frame, water maze + tracking software, set-shift and maze apparatus, cryostat, fluorescence microscope and image-analysis software, western blot system. Hourly facility charges are NOT in any scenario; without written access the budget is void.
Personnel, services, overhead	Not in the budget. Ten operators (two shifts of five), a surgeon and a coordinator for ≈6–7 months including Phase 0a, plus institutional overhead (typically 15–30%, institute-specific), must be quoted by the host. The USD 10.2k figure is consumables-only.
Timeline	IAEC 1–6 months; FGF17 procurement is serial — pack 1 import (6–12 weeks) → bioassay (≈2 weeks) → pack 2 import (6–12 weeks) ≈ 3–6 months before the main cohort can start; other procurement runs in parallel; Phase 0a 1–2 months; in-life ≈ 4.9 months (21 weeks); analysis 1–2 months. About 9–16 months from IAEC submission to core results; host search, contracting and FCRA/GST arrangements come before that and are not counted.

Budget (USD; list prices from search plus reviewer knowledge; to be replaced by quotes):

Line (core = Phase 0a + 0b)			Base	Notes
Rats: 136 incl. satellites, partners, pilots, with transport			1.1k	INR 300–500 each
Housing/care ≈ 77 days per main-cohort rat; partners longer			1.5k	Indian anchor (unverified, reviewer search): NCBS ≈ INR 41–44 (subsidised) / 81–82 (non-subsidised) per rat cage per day; optimistic 1.0k
FGF17 ≈100 µg (2 × 50 µg packs; a 25 µg vial is the contingency), landed			1.3k	Cohesion ≈ USD 480/50 µg via an aggregator; R&D; 25 µg mouse vial price not found; PeproTech lists human only; a 3rd vial (contingency) 0.45k in pessimistic
7,8-DHF ≈ 4 g, landed			0.35k	TCI ≈ USD 62/g
Cannulas, dummies, screws, cement (100 rats)			0.55k	
Anaesthesia, analgesia, surgical and terminal consumables (≈100 cannulations, ≈136 terminal procedures)			1.05k	
ELE arena, wheels, objects, puzzle devices (18% GST on fabrication)			0.65k	
Behaviour consumables and fabrication (set-shift pots/media/odours, Hebb-Williams maze) incl. GST			0.8k	
EdU (banks the label)			0.2k	
ERK block, landed: pERK, total ERK, HRP secondaries, loading control, FGF2 positive control			1.55k	list ≈ 1.15k
Western reagents (membranes, ECL, inhibitors)			0.45k	
Two shipments with customs-agent fees			0.6k	
IAEC fee and miscellaneous			0.1k	
Core base total, consumables only, no reserve			10.2k	excludes personnel, overhead, equipment fees
Phase 0c add-on (deferred): EdU click kit + Olig2 + secondary landed 1.34k, IHC reagents 0.25k, shipping 0.1k			1.7k	CC1 (≈USD 594 list) and PSD-95/synaptophysin westerns are further add-ons
Full programme base			11.9k	
Scenario	Core	Full	Assumptions (itemised)	
Optimistic	≈ 8.9k	≈ 10.3k	minus housing 0.5k, FGF17 0.2k, antibodies at list 0.4k, one cheaper shipment 0.2k (core); 0c at list ≈1.35k	
Base	≈ 10.2k	≈ 11.9k	as above	
Pessimistic	≈ 12.2k	≈ 14.1k	15% overrun plus a contingency 25 µg FGF17 vial (full programme: 0c also +15%); equipment fees, personnel and overhead still excluded (the base already assumes no DSIR concession)	

Ceiling (researcher-approved): USD 12k for materials. Core base 10.2k and core + 0c base 11.9k are inside it; the pessimistic core (12.2k) slightly exceeds it and the pessimistic full programme (14.1k) exceeds it, so 0c runs last and is dropped first. **Phases:** 0a ≈ 4.1k (satellite rats, first FGF17 vial, ERK antibodies, surgery, first shipment) with a stop rule at that spend; 0b ≈ 6.1k; 0c ≈ 1.7k deferred. **First cuts if prices run high:** enrol 14 per arm (waves of 7 + 7; feasibility gate becomes ≥28); drop the 60-min satellite group; run 0c later. The western block must not be dropped (it carries the only gate). **Must not be cut:** dye/histology placement check, FGF17 bioassay and alignment, blinding/randomisation, DHF tolerability check, visible-platform control, banking of tissue.

10. Lab handover and remote oversight

The researcher will not be on site; the host laboratory runs the experiment. This PDF is the working protocol; the following must exist before animals arrive.

Item	Responsibility / requirement
Legal form (decide first)	Either (a) an academic collaboration under an MoU — then the institute needs FCRA registration or prior permission and a designated account for foreign contributions (likely required for foreign research funds, received in a designated bank account; payment for services rendered is outside FCRA; confirm in writing with the institute's accounts office; the researcher's nationality is stated in the agreement because it may change the treatment); or (b) a fee-for-service contract — invoice, GST position or export-of-services status and TDS must be agreed. The choice determines payment mechanics and is a precondition for submission.

Item	Responsibility / requirement
Host selection and due diligence	Criteria: CCSEA/CPCSEA-registered establishment with a functioning IAEC; written record of ICV cannulation, set-shifting and water-maze work; access to all listed equipment; a named attending veterinarian; references from two previous external collaborations. Trade-off: a CRO offers fee-for-service and quality systems but rarely these assays; an academic lab offers expertise but expects authorship and slower timelines.
Payment tied to gates	Tranches: (1) at signature (set-up, animal order); (2) after the host-competence gate; (3) after Phase 0a passes; (4) at data lock. A failed 0a stops spending at its tranche. The agreement states who bears currency risk.
Governance and welfare oversight	Protocol amendments need the host PI, the sponsor and (where required) the IAEC; a named attending veterinarian can stop the study; weekly remote welfare report (weights, scores, cap losses, adverse events); escalation: sponsor within 48 h, institute head within 7 days; working language English; a fixed weekly call slot overlapping both time zones.
Quality standard	Non-GLP; the host works to GLP-like good practice (contemporaneous records, calibrated equipment, training records, retained raw data).
Roles and accountability	Host principal investigator (accountable for IAEC compliance, animal welfare; also the IAEC PI); study coordinator (one named person); ten trained operators in two shifts of five plus a surgeon and a weekend rota; sponsor (design, randomisation, drug-code key, registration, independent analysis).
Agreement terms	Funding, overhead and payment schedule; who pays for overruns and for a failed Phase 0a; IP ownership; data and tissue ownership; insurance and liability; stop rules; publication and authorship (ICMJE criteria); data sharing.
Host-competence gate (before Phase 0a)	Practice-group evidence: $\geq 80\%$ successful ICV cannulation by dye (if the gate fails, the sponsor funds a short training and re-test or reassigns the host, and no main-cohort money is spent until it is met); western blot CV from the lab's previous work; set-shifting and water-maze pilot records; verified equipment list. Sponsor decides go/no-go on this evidence.
SOP pack (host completes with photos; sponsor reviews)	Surgery and cannula care; ICV bolus injection; vehicle preparation and dosing; ELE and control sessions; set-shifting; Hebb-Williams; water maze incl. temperature/hypothermia rules; food restriction and weighing; euthanasia, hemisection, slab fixation and tissue banking; satellite dissection and western protocol; per-day time-budget table; welfare scoresheet; EdU-injection SOP; data-recording and imaging SOP; Appendix A parameters signed off. The ethics (IAEC) application is drafted jointly: the sponsor supplies a model narrative (severity, endpoints, justification), the host adapts it. The sponsor supplies spreadsheet templates: daily log (weights, doses, rations), session logs (ELE/control), behaviour score sheets, and a one-page weekly report.
Blinding and randomisation	Sponsor generates the allocation, labels coded vials and arm IDs, and keeps the key; host staff handle coded IDs only; unblinding after data lock.
Data integrity and monitoring	Bound or electronic lab notebooks with audit trail; raw files hashed on upload; protocol deviations classed minor/major with major ones reported within 48 h and a sponsor decision; scheduled live-video audits and source-data verification of $\approx 10\%$ of records; weekly summary (enrolment, weights, exclusions, adverse events, cap losses) and photo/video check-in; raw data and images shared with the sponsor for independent blinded analysis.
Tissue and image transfer	Banked tissue stays at the host unless a material transfer agreement and the necessary export permits are in place; image format and storage (coded IDs, backup) agreed in advance.
Reagent chain	Quotes agreed; vial-first FGF17 order; bioassay result and alignment shared before the main cohort starts; cold-chain receipt log.
Decisions	After 0a and 0b the sponsor issues a written decision (provisional go / stop / redesign) before further spending.

Appendix A. Task parameters (defaults; host to confirm and sign off before animals arrive)

Task	Parameters
Set-shifting (digging; Birrell & Brown 2000; Tait et al. 2018 [11,19])	Two ceramic pots per trial; stimulus dimensions: digging medium and odour (neither is used in the ELE); 2–3 days digging habituation to a baited pot; example exemplar sets (host to fix ≥ 6 each before animals arrive): odours — cinnamon, nutmeg, clove, cumin, thyme, vanilla; media — sawdust, shredded paper, sand, plastic beads, wood chips, cotton; stages: simple discrimination, compound discrimination, first intradimensional shift (ID1), second intradimensional shift (ID2, to build the attentional set), extradimensional shift (reversal stages omitted to limit session length — declared); validity check: ED trials > ID2 trials cohort-wide; dimension order and exemplar sets counterbalanced across rats and arms; odour applied to the medium and bait buried in both pots (masked), with a fixed discovery-trial rule; criterion 6 consecutive correct; maximum 60 trials per stage; 3-min trial limit; error on first incorrect dig; bait $\approx \frac{1}{2}$ cereal piece; pot positions pseudo-random; training cap 7 days.
Hebb-Williams (Rabinovitch & Rosvold 1951 [22])	Closed-field maze with barrier layouts; 3 days adaptation with food reward including the standard practice problems; 6 test problems chosen from the standard 12 by host and sponsor before animals arrive (fixed order, same layouts for all rats; investigator-chosen subset, not validated); 6 trials per problem (3-min cutoff), one problem per day, two identical mazes in parallel; error zones defined in advance; split-half reliability reported.
Morris water maze	Pool ≈ 150 – 180 cm, 25 ± 1 °C, hidden 10-cm platform 1.5–2 cm below the surface in opaque water (non-toxic black tempera paint, for contrast with albino rats), one pool, ≥ 4 distal cues, dim white light (≈ 10 – 20 lux, measured; spectrum recorded) in the pool room only — a declared exception to the dark-phase red-light rule; rats travel in covered cages and the colony room stays dark; 4 trials/day, 60 s maximum, four quasi-random start points, ≥ 5 min between trials in a warm cage (32 °C); core temperature before and after sessions (rest day if < 34.5 °C); 5 acquisition days; 60-s probe 24 h after the last acquisition day; visible-platform test (4 trials) after the probe; scores: mean proximity (Gallagher), target-quadrant time, swim speed, floating, thigmotaxis.
ICV cannula (to be confirmed in the practice group)	Right lateral ventricle, rat aged 8–10 weeks (≈ 250 – 300 g): from bregma AP -0.9 mm, ML $+1.5$ mm, DV such that the injector tip lies at -4.0 to -4.5 mm (the host states whether DV is measured from bregma or the skull surface; guide ≈ -3.0 to -3.5 mm with the injector protruding ≈ 1 mm), scaled by the bregma-lambda distance and confirmed by dye in the practice group; fixed with screws and dental cement under a protected recessed cap; dummy cannula between injections.
FGF17 bioassay (0a reagent gate)	Serum-starved FGF-responsive cells (the supplier's BALB/3T3 line, or NIH-3T3; primary rat OPCs if available) stimulated for 10–15 min with 0–100 ng/ml of the supplied FGF17 in the presence of heparin 10 $\mu\text{g/ml}$ (an EC50 comparison is only valid with matching heparin); pERK/total ERK by western; FGF2 as positive control; pass = a clear dose-response with ≥ 3 -fold maximal pERK over unstimulated cells; the supplier's stated ED50 is a proliferation value, so on first purchase the pERK dose-response is judged against FGF2 and the ≥ 3 -fold criterion (an earlier-lot comparison is only possible for later lots). A 3T3 pass confirms potency through FGFR1c/2c, not OPC-relevant FGFR3/4 signalling.
Antibodies and kits (host to confirm; unverified examples from memory)	pERK1/2 Thr202/Tyr204 (e.g. CST #4370), total ERK1/2 (e.g. CST #4695), Olig2 (e.g. Millipore AB9610), EdU click kit (e.g. Thermo Click-iT EdU imaging kit); the host lists final catalogue numbers, rat reactivity and dilutions in the SOP pack.
ELE / control session	Dark phase under red light; pair session of 60 min (test rat + partner); 4 arenas; arms balanced within every slot (one rat per arm per slot), no rotation; log object list, puzzle level, rewards, engagement minutes, wheel revolutions.

Appendix B. Operator time budget per wave (to be confirmed by the host)

Period	Daily work	Operator-hours/day
Days -14 to -11	Cannula surgery, 8 rats/day (surgeon + assistant)	≈ 16
Days 0-52, every day incl. weekends (routine)	ELE/control: 32 pair-sessions in 4 arenas (≈9 h per arena with changeovers), 1 operator per 2 arenas ≈ 18 h; DHF injections ≈ 1.5 h; weighing, ration and cage checks ≈ 1.5 h; cleaning and partner care ≈ 1.5 h; video scoring and logs ≈ 1 h	≈ 24
Days 0-6 (extra)	ICV boluses (32 rats ≈ 2 h × 2 operators) and EdU	+ 4
Days 18-27 (set-shift habituation and training)	32 rats × ≈1 h, each in its own slot	+ 32 (total ≈ 56)
Days 28-36 (set-shift tests)	8 rats in testing per day (4 on day 1, 4 on day 2) × up to 3 h, one operator per rat, in their own slots	+ 24 (total ≈ 48)
Days 37-45 (Hebb-Williams)	32 rats × 6 trials in two mazes ≈ 8 h per maze	+ 16 (total ≈ 40)
Days 46-52 (water maze)	32 rats × 4 trials, ≥5 min between trials, two core temperatures ≈ 9 h	+ 9 (total ≈ 33)
Days 53-56 (euthanasia)	8 rats/day × ≤12 min with two operators plus tissue handling; no routine (ELE and dosing end on day 52)	≈ 8
Weekends	Routine only	≈ 24

The peak is ≈ 56 operator-hours a day (set-shift training), against ten operators × 8 h = 80 across two shifts of five (peak concurrency at a slot crossover is ≈6-8, so five per shift is tight); weekly need ≈ 250-300 operator-hours. The surgeon is additional on surgery days. Personnel cost is not in the budget.

11. Validation record for this adaptation

Round	Reviewers	Verdicts	Main issues → fix
Earlier A	3 Opus (design; pharmacology; logistics)	All approve with changes	Pump ends before testing; pump cost; middle-aged rats; tissue conflict → boluses, young adults, new sourcing.
Earlier B	3 Opus + 1 Haiku; the aim was changed	All approve with changes	Transfer overlap; social confound; composite not a g factor; attrition; timeline; budget.
Translational	1 Opus	Approve with changes	Overclaiming; neurotrophin history; FGF safety; paediatric ethics.
Round 1	5 Opus + 2 Sonnet	All 7 approve with changes	Gate miscalibrated; BrdU/Olig2 on adjacent sections; phospho pre-analytics; timelines; ELE trained set-shift dimensions; budget overhead; 25 inconsistencies.
Round 2	5 Opus + 2 Sonnet	All 7 approve with changes	“Go” rule not a gate; aliased covariates; complete-case bias; no frozen tissue; n=4 satellites; transfer mislabelled; Hebb-Williams reliability; wave workload; shipping vs vial-first; 17 inconsistencies.
Round 3	5 Opus + 2 Sonnet	All 7 approve with changes; 8 objections called blocking (all fixable on paper)	Whole-hemisphere immersion fixation fails for deep regions → slab fixation, contralateral hemisphere; corn-oil IP x53 → CMC-Na default + necropsy scoring (CMC-Na default superseded in round 4); overlapping waves impossible in 4 arenas → sequential waves; <7 days surgical recovery → surgery D-14; no staff/overhead costing; legal form, IAEC PI, IP and insurance; data-integrity regime; host-competence gate; wave as fixed effect; set-shift scoring; restriction vs humane endpoint; time-of-day; parameter appendix; SD conversion error; 8 inconsistencies → all applied in v4.
Round 4	5 Opus + 2 Sonnet (one Opus review ended on a safeguard flag after delivering its report)	All approve with changes; blocking: cannula placement vs frozen hemisphere, host executability, gate-linked payments, CMC-Na exposure, FGF17 gate anchoring, ED-on-ID overadjustment, gate power, ELE-correlated cap loss	Dye + fresh slab for placement; low-DMSO solution default and extra DHF time point; FGF17 receptor rationale, gate power and re-test; raw log-ED primary; protected caps and attrition stop; DHF before ELE; Appendix B time budget; host selection, payment tranches, governance; 1.5-2 mm slabs; cross-gel bridging control; in-life 21 weeks; 136 rats.
Round 5	5 Opus + 2 Sonnet	All approve with changes; blocking: session window/operator load, white pool paint, set-shift validity (ID2), undefined FGF17 bioassay, missing cannula coordinates	Fixed daily slots; six operators in two shifts; black tempera; ID2 stage and counterbalancing; bioassay and coordinates added; Iram dose anchored (0.5 µg × 4); vehicle check extended to 28 days; wave 2 day 0 = wave 1 day 71; Fortin wording corrected; customs wording corrected; geometry, gel-blocking and EdU interpretation declared.
Round 6	5 Opus + 2 Sonnet	All approve with changes (behaviour: major revision of the schedule); blocking: fixed slots vs Latin square, set-shift test length, staffing deficit, two-stage gate, circular threshold, set-shift censoring, ICV-failure contingency	One rat per arm in every slot; set-shifting split over two days in own slot; eight operators; composite gate performance; raw-unit threshold frozen from wave 1; censored analysis primary if >20% at cap; micropunch and periventricular labelling; reference renumbering; many wording fixes.
Round 7	5 Opus + 2 Sonnet	All approve with changes; behaviour: not yet executable (set-shift test pipeline, staffing, dark-phase arithmetic, equipment)	Set-shift moved earlier (habituation day 18, tests days 28-35) with 8 concurrent testers and ≥8 boxes; ten operators in two shifts of five; time-matched vehicle in FGF17 satellites (30 rats; total 136; budget core ≈10.2k, ≈12.2k pessimistic); micropunch claim corrected and yield measured; interface wording for the gate; tolerability with a 28-day interim gate and 53-day follow-up; serial FGF17 import path (9-16 months); frozen-SD rule and coded-group unblinding; Cox primary; slot as random block; vendor concentration claim corrected; refs 11/22/23; many wording fixes.

Round	Reviewers	Verdicts	Main issues → fix
Round 8	5 Opus + 2 Sonnet	All approve with changes; no arithmetic errors remain; blocking-on-paper: stale 8.7k header, “just reaches” wording, gate simulated at n=6 vs actual 4 vs 6, minimal ELE specification, engagement-gate specificity control, 28- vs 53-day tolerability wording	Header 8.9k; pessimistic core “slightly exceeds” 12k; gate re-simulated at 4 vs 6 (false-pass ≈6% / 30%); punch laterality and diluted-tissue threshold stated; customs wording updated (Notification 45/2025-Customs, 1 Nov 2025, consolidating 51/96); heat-inactivated FGF17 control; tolerability day-28/53 wording; BSA labelled; 3rd-vial wording; image-analysis software; 0.6× wording; remaining items listed in section 12.

12. Items the reviewers still do not like / remaining gaps (after round 8, final)

#	Gap	Kind
1	Bolus FGF17 is not the Iram infusion (≈7× less volume, pulsatile); the dose now matches Iram’s reported ≈2.1 µg, but that figure rests on one reviewer’s retrieval; a null remains confounded with delivery.	Pumps in the larger study
2	Power: only very large effects detectable; healthy rats near ceiling; the direction rule is lenient and cannot separate synergy from additivity (38% go for enrichment alone).	Intrinsic to a pilot
3	7,8-DHF may be inert in vivo; 53 injections at the long end of precedent; pERK shows pathway engagement only; DHF gate descriptive.	Declared
4	ELE confounds social contact, exercise, puzzles and spatial experience; weak 60-min dose; arm cannot be fully blinded; set-shifting is the only clean transfer readout.	Declared
5	Cumulative welfare burden may be challenged by the IAEC; attrition may exceed the plan; staffing (ten operators in two shifts of five, surgeon, weekend rota) to be confirmed by the host.	Pre-review; host
6	Prices unverified; personnel, overhead and equipment fees excluded; base core ≈10.2k and base full ≈11.9k are inside the researcher-approved USD 12k ceiling; the pessimistic core (12.2k) slightly exceeds it and the pessimistic full programme (14.1k) exceeds it; DSIR reagent coverage and FCRA/GST status unconfirmed.	Quotes and written confirmations
7	Structure (EdU/Olig2) is deferred and is read on the left hemisphere, opposite the injected ventricle (a near-ventricle asymmetric effect would be missed), so the provisional go lacks a structural marker; immersion-fixed deep-region histology is lower quality than perfusion; no synaptic or plasticity readout in this study.	Phase 0c / Stage 1a
8	Optogenetics (Stage 1b) and the human roadmap are aspirations without design detail or validated tools in the host lab.	Own review round
9	Round-8 reviewers added: the engagement gate has no FGFR-inhibitor arm (only a heat-inactivated control); the ELE is only minimally specified and delegated to the host; ten operators in two shifts of five plus a surgeon may exceed a typical Indian academic facility and personnel/overhead are excluded from the USD 12k headline (the dominant real cost); gate false-pass is higher than first simulated; Hebb-Williams subset and set-shift ID2 are investigator-chosen and not validated; the customs notification must be confirmed with a customs agent. Literature checks used search snippets only; Iram’s delivered dose was reported by one reviewer only; human-trial and neurotrophin statements to be verified.	Researcher/host to verify
10	Vehicle choice may confound exposure (suspension fallback); 7,8-DHF is at trough at each day’s testing; wave and calendar are confounded (arm × wave not estimable).	Declared
11	Host executability depends on SOPs, quotes and written confirmations (FCRA/GST, DSIR, equipment access) that are not yet in hand.	Host to supply

13. References and verification status

Checks were by search snippets only (PubMed, PMC, Crossref and publisher pages unreachable). “Verified” means consistent across indexed records.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. Nature 2022;605:509-515. doi:10.1038/s41586-022-04722-0 (an Author Correction exists)	Verified; delivered dose (25 µg/ml; 0.5 µl/h × 7 d ≈ 2.1 µg) reported by one reviewer, not independently confirmed
2	Boltaev U, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. Sci Signal 2017;10(493). doi:10.1126/scisignal.aal1670	Verified

#	Citation	Status
3	Todd D, Gowers I, Dowler SJ, et al. PLoS One 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; DHF content via reviews
4	Do small molecules activate the TrkB receptor in the same manner as BDNF? Pharmaceuticals 2021;14(8):704 (authors not captured).	Verified (review)
5	Rawlings-Mortimer F, et al. 7,8-DHF during water-maze training in mice. Front Syst Neurosci 2023. doi:10.3389/fnsys.2023.1134594	Partial
6	van Praag H, Kempermann G, Gage FH. Nat Rev Neurosci 2000;1:191-198. doi:10.1038/35044558	Verified
7	Nithianantharajah J, Hannan AJ. Nat Rev Neurosci 2006;7:697-709. doi:10.1038/nrn1970	Verified
8	Galsworthy MJ, et al. Behav Genet 2005;35:675-692. doi:10.1007/s10519-005-3423-9	Verified
9	Matzel LD, Han YR, Grossman H, et al. J Neurosci 2003 (general learning ability in mice)	Partial
10	Gibson EM, Purger D, Mount CW, et al. Science 2014;344:1252304 (mouse optogenetic premotor-cortex stimulation promotes oligodendrogenesis and myelination)	Verified
11	Birrell JM, Brown VJ. Medial frontal cortex mediates perceptual attentional set shifting in the rat. J Neurosci 2000;20(11):4320-4324 (published protocols differ in stages; some add ID2, which is used here as a modification)	Verified
12	Kraemer HC, et al. Arch Gen Psychiatry 2006;63:484-489. doi:10.1001/archpsyc.63.5.484	Verified
13	Leon AC, Davis LL, Kraemer HC. J Psychiatr Res 2011;45:626-629. doi:10.1016/j.jpsychires.2010.10.008	Verified
14	Percie du Sert N, et al. PLoS Biol 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified
15	Enrichment meta-analysis (rodent cognition $g \approx 0.44$, 95% CI 0.25-0.62): Univ. at Albany dissertation (scholarsarchive.library.albany.edu/etd/371); Léger M, et al. 2014 (duration of enrichment effects in mice)	Partial; dissertation not peer-reviewed (Léger 2014 is peer-reviewed; citation not fully retrieved)
16	Fortin D, Rom E, Sun H, Yayon A, Bansal R. Distinct FGF/FGF receptor signaling pairs initiate diverse cellular responses in the oligodendrocyte lineage. J Neurosci 2005;25:7470-7479. doi:10.1523/JNEUROSCI.2120-05.2005	Verified
17	Nie et al. 7,8-DHF protects nigrostriatal dopaminergic neurons from rotenone-induced neurotoxicity in rodents. Rotenone model in 9-month-old rats, 7,8-DHF 5 mg/kg IP daily for 5 weeks. Parkinsons Dis 2019, article 9193534 (PMC6421741; authors not captured)	Verified (snippet)
18	Optic nerve transection in rats, 7,8-DHF 5 mg/kg IP daily in 1% DMSO/saline, outcomes at 7, 14 and 21 days (Gallego-Ortega et al., Int J Mol Sci 2021;22:11815; PMC8584116)	Partial
19	Tait DS, Bowman EM, Neuwirth LS, Brown VJ. Assessment of intradimensional/extradimensional attentional set-shifting in rats. Neurosci Biobehav Rev 2018;89:72-84	Verified
20	Browne RH. On the use of a pilot sample for sample size determination. Stat Med 1995;14:1933-1940	Verified (title to check)
21	Gallagher M, Burwell R, Burchinal M. Severity of spatial learning impairment in aging: development of a learning index for performance in the Morris water maze. Behav Neurosci 1993;107:618-626. doi:10.1037/0735-7044.107.4.618	Verified
22	Rabinovitch MS, Rosvold HE. A closed field intelligence test for rats. Can J Psychol 1951;5(3):122-128 (issue number unconfirmed; Hebb-Williams maze)	Partial
23	Zhang X, et al. Receptor specificity of the fibroblast growth factor family: the complete mammalian FGF family. J Biol Chem 2006;281:15694-15700 (FGF8 subfamily expected to act through FGFR3c/FGFR4; FGF17-specific profile not retrieved)	Real; FGF17 result unverified

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young rodents; any study combining 7,8-DHF with enrichment; rat ICV FGF17 doses; independent confirmation of Iram 2022's delivered dose; Indian prices for Alzet pumps and recombinant FGF17; a peer-reviewed validation of a Hebb-Williams subset; per-diem housing cost.